



Information Technology University, Lahore

Applied Machine Learning

BSCE- 23 Spring-2026

Assignment # 01

Issue Date: 26/02/2025

Due Date: 03/02/2025

Total marks =100

Instructions

1. Please review the University Plagiarism Policy.
2. Any submissions handed in late will not be accepted and will receive a zero score.
3. This Assignment will access your CLOs as per OBE.
4. Assignment should be uploaded as PDF file with Notebook in Google Classroom.
5. The name of the file should be your Roll Number as; BSCEXXXX.
6. Please submit your own work only.

Instructions for coding part:

- Google Colab (Jupyter Notebook) is recommended for the implementation of Models.

Question # 01: Linear Regression with Custom Dataset (50 Marks)

Problem Statement

You are hired as a Data Analyst in a marketing company. The company wants to predict product sales based on advertising budgets allocated across different platforms.

Your task is to generate a custom dataset and build a Linear Regression model to predict sales.

Dataset Requirements

Create a customized dataset named "Marketing_Sales" containing 500 randomly generated records with the following columns:

- Campaign_ID
- TV_Budget
- Social_Media_Budget
- Newspaper_Budget
- Sales (Target Variable)

You may generate random realistic values for budgets and compute Sales using a linear formula with some random noise.

Specific Tasks

1. Data Generation & Preparation

- Generate dataset using NumPy / Pandas
- Display first 5 rows
- Identify:

- Independent Variables (X)
- Dependent Variable (Y)

2. Train-Test Split

- Split dataset into training and testing sets using 70:30 ratio
- Print:
 - X_train
 - X_test
 - y_train
 - y_test

3. Model Implementation

- Build a Multiple Linear Regression model using training data

4. Model Evaluation

Print the following:

- Intercept
- Coefficients
- R² Score

Also calculate:

- Mean Absolute Error (MAE)
- Mean Squared Error (MSE)

5. Prediction & Analysis

- Predict sales for the test dataset
- Display comparison of Actual vs Predicted Sales
- Plot a graph showing Actual vs Predicted values

6. Interpretation

- Which marketing channel contributes most to sales?
- Interpret coefficients in practical terms.

Question # 02: (50 Marks)

Problem Statement:

You are a student at research institute. The institute has collected measurements of iris flowers but needs help automatically classifying them into species based on their physical characteristics. Your task is to build and compare multiple classification models to accurately predict iris species. Given a dataset of iris flower measurements, develop and evaluate three different classification algorithms to predict the species of iris flowers based on their sepal and petal dimensions.

The Dataset

The famous Iris dataset contains 150 samples with 4 features:

- Sepal length (cm)
- Sepal width (cm)

- Petal length (cm)
- Petal width (cm)
- Target variable: Species (Setosa, Versicolor, Virginica)

Your Task Requirements:

Implement and compare three classification methods:

- **K-Nearest Neighbors (KNN)**
- **Logistic Regression**
- **Decision Tree Classifier**

Specific Tasks:

1. Data Preparation:

- Load and explore the dataset
- Split into training and testing sets
- Perform any necessary preprocessing

2. Model Implementation:

Implement all three classifiers with appropriate parameters:

KNN: Experiment with different k values (3, 5, 7)

Logistic Regression: Use appropriate solver and maximum iterations

Decision Tree: Experiment with different max_depth values

3. Model Evaluation

For each model, calculate:

- Accuracy score
- Confusion matrix
- Classification report (precision, recall, f1-score)

4. Comparative Analysis:

Compare the performance of all three models and answer:

1. Which algorithm performed best?
2. What are the strengths and weaknesses of each method?
3. Which model would you recommend for production and why?

Expected Deliverables:

- Make Report with results, comparison
- Complete Python code implementing in the python file containing all models.
- Performance comparison table showing metrics for each model
- Visualizations: Confusion matrices for each model
- Conclusion report with recommendations